Variation of Parameters for Systems

For $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{f}(t)$, the **fundamental matrix** $\boldsymbol{\Phi}$ has independent complementary solutions as its columns, so $\boldsymbol{\Phi}' = \boldsymbol{A}\boldsymbol{\Phi}$.

Replace the constant vector in $\boldsymbol{y}_c = \boldsymbol{\Phi}\boldsymbol{c}$ by $\boldsymbol{u}(t)$. Substituting $\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{u}$ gives

$$\boldsymbol{\Phi}(t)\boldsymbol{u}'(t) = \boldsymbol{f}(t) \qquad \Longrightarrow \qquad \boldsymbol{u}'(t) = \boldsymbol{\Phi}(t)^{-1}\boldsymbol{f}(t)$$

One particular solution is

$$\boxed{\boldsymbol{y}_p(t) = \boldsymbol{\Phi}(t)\int_{t_0}^{t} \boldsymbol{\Phi}(s)^{-1}\boldsymbol{f}(s)\,ds}$$

where t_0 is any point in the interval of the system.

Worked Example

$\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \boldsymbol{y} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Find a particular solution of $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{f}(t)$ on $(0, \infty)$, where

$$\boldsymbol{A}(t) = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \qquad \boldsymbol{f}(t) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The homogeneous system $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y}$ has independent solutions

$$\boldsymbol{y}_1(t) = \begin{pmatrix} t^2 \\ 2t \end{pmatrix} \qquad \boldsymbol{y}_2(t) = \begin{pmatrix} t^3 \\ 3t^2 \end{pmatrix}$$

Verify Homogeneous Solutions

$\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \boldsymbol{y} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

For the homogeneous equation $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y}$, check both vectors on $t > 0$.

$$\boldsymbol{A}(t)\boldsymbol{y}_1(t) = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \begin{pmatrix} t^2 \\ 2t \end{pmatrix} = \begin{pmatrix} 2t \\ -6+8 \end{pmatrix} = \begin{pmatrix} 2t \\ 2 \end{pmatrix} = \boldsymbol{y}_1'(t)$$

$$\boldsymbol{A}(t)\boldsymbol{y}_2(t) = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \begin{pmatrix} t^3 \\ 3t^2 \end{pmatrix} = \begin{pmatrix} 3t^2 \\ -6t+12t \end{pmatrix} = \begin{pmatrix} 3t^2 \\ 6t \end{pmatrix} = \boldsymbol{y}_2'(t)$$

Fundamental Matrix

$$\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \boldsymbol{y} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Use the given homogeneous solutions as columns.

$$\boldsymbol{\Phi}(t) = \begin{bmatrix} \boldsymbol{y}_1(t) & \boldsymbol{y}_2(t) \end{bmatrix} = \begin{pmatrix} t^2 & t^3 \\ 2t & 3t^2 \end{pmatrix}$$

$$\det \boldsymbol{\Phi}(t) = 3t^4 - 2t^4 = t^4 > 0 \qquad \text{for } t > 0$$

Thus $\boldsymbol{\Phi}(t)$ is invertible throughout $(0, \infty)$.

Fundamental Matrix Inverse

$$\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \boldsymbol{y} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

For an invertible 2×2 matrix, swap the two diagonal entries, negate the two off-diagonal entries, and divide the resulting matrix by the determinant.

$$\boldsymbol{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \boldsymbol{M}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

The determinant of $\boldsymbol{\Phi}(t)$ is t^4, so

$$\boldsymbol{\Phi}(t) = \begin{pmatrix} t^2 & t^3 \\ 2t & 3t^2 \end{pmatrix} \qquad \boldsymbol{\Phi}(t)^{-1} = \frac{1}{t^4} \begin{pmatrix} 3t^2 & -t^3 \\ -2t & t^2 \end{pmatrix}$$

Variation of Parameters

$\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \boldsymbol{y} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Write $\boldsymbol{y} = \boldsymbol{\Phi u}$ and multiply

$$\boldsymbol{u}'(t) = \Phi(t)^{-1}\boldsymbol{f}(t) = \frac{1}{t^4}\begin{pmatrix} 3t^2 & -t^3 \\ -2t & t^2 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\boldsymbol{u}'(t) = \frac{1}{t^4}\begin{pmatrix} -t^3 \\ t^2 \end{pmatrix} = \begin{pmatrix} -t^3/t^4 \\ t^2/t^4 \end{pmatrix} = \begin{pmatrix} -t^{-1} \\ t^{-2} \end{pmatrix}$$

$$\boldsymbol{u}'(t) = \begin{pmatrix} -t^{-1} \\ t^{-2} \end{pmatrix} \qquad \Longrightarrow \qquad \boldsymbol{u}(t) = \int \begin{pmatrix} -t^{-1} \\ t^{-2} \end{pmatrix} dt = \begin{pmatrix} -\log(t) \\ -1/t \end{pmatrix} + \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

General Solution

$$\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \boldsymbol{y} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Multiply by $\boldsymbol{\Phi}(t)$ and distribute.

$$\boldsymbol{y}(t) = \boldsymbol{\Phi}(t)\boldsymbol{u}(t) = \boldsymbol{\Phi}(t)\left[\begin{pmatrix} -\log(t) \\ -1/t \end{pmatrix} + \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}\right]$$

$$\boldsymbol{y}(t) = \begin{pmatrix} t^2 & t^3 \\ 2t & 3t^2 \end{pmatrix}\begin{pmatrix} -\log(t) \\ -1/t \end{pmatrix} + \begin{pmatrix} t^2 & t^3 \\ 2t & 3t^2 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\boldsymbol{y}(t) = \underbrace{\begin{pmatrix} -t^2\log(t) - t^2 \\ -2t\log(t) - 3t \end{pmatrix}}_{\text{particular solution } \boldsymbol{y}_p(t)} + \underbrace{c_1\boldsymbol{y}_1(t) + c_2\boldsymbol{y}_2(t)}_{\text{complementary solution}}$$

Verification

$\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -6/t^2 & 4/t \end{pmatrix} \boldsymbol{y} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Differentiate the two components of $\boldsymbol{y}_p$.

$$\boldsymbol{y}_p'(t) = \begin{pmatrix} -2t\log(t) - 3t \\ -2\log(t) - 5 \end{pmatrix}$$

The right-hand side of the system gives the same vector.

$$\begin{aligned} \boldsymbol{A}(t)\boldsymbol{y}_p(t) + \boldsymbol{f}(t) &= \begin{pmatrix} -2t\log(t) - 3t \\ 6\log(t) + 6 - 8\log(t) - 12 + 1 \end{pmatrix} \\ &= \begin{pmatrix} -2t\log(t) - 3t \\ -2\log(t) - 5 \end{pmatrix} = \boldsymbol{y}_p'(t) \end{aligned}$$

Constant-Coefficient Complex Eigenvalues

$\boldsymbol{y}' = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)\boldsymbol{y} + \left(\begin{smallmatrix} t \\ 0 \end{smallmatrix}\right)$

Find the solution of the first-order initial value problem

$$\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y} + \boldsymbol{f}(t) \qquad \boldsymbol{y}(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

where

$$\boldsymbol{A} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} \qquad \boldsymbol{f}(t) = \begin{pmatrix} t \\ 0 \end{pmatrix}$$

The characteristic polynomial of $\boldsymbol{A}$ is

$$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \det \begin{pmatrix} \lambda & 2 \\ -2 & \lambda \end{pmatrix} = \lambda^2 + 4$$

The eigenvalues are $2i$ and $-2i$.

Eigenvector for $\lambda = +2i$

$\boldsymbol{y}' = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)\boldsymbol{y} + \left(\begin{smallmatrix} t \\ 0 \end{smallmatrix}\right)$

The first row of $(\boldsymbol{A} - 2i\boldsymbol{I})\boldsymbol{v} = \boldsymbol{0}$ gives $-2iv_1 - 2v_2 = 0$, so $v_2 = -iv_1$. Choose $v_1 = 1$.

$$(\boldsymbol{A} - 2i\boldsymbol{I})\begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} -2i & -2 \\ 2 & -2i \end{pmatrix}\begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The resulting complex homogeneous solution is

$$\boldsymbol{z}(t) = e^{2it}\boldsymbol{v} = \big(\cos(2t) + i\sin(2t)\big)\begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} \cos(2t) \\ \sin(2t) \end{pmatrix} + i\begin{pmatrix} \sin(2t) \\ -\cos(2t) \end{pmatrix}$$

Real y_c

$$\boldsymbol{y}' = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)\boldsymbol{y} + \left(\begin{smallmatrix} t \\ 0 \end{smallmatrix}\right)$$

Take $\boldsymbol{y}_1(t) = \operatorname{Re}(\boldsymbol{z}(t))$ and $\boldsymbol{y}_2(t) = -\operatorname{Im}(\boldsymbol{z}(t))$ to form $\boldsymbol{\Phi}(t)$.

$$\boldsymbol{\Phi}(t) = \begin{bmatrix} \boldsymbol{y}_1(t) & \boldsymbol{y}_2(t) \end{bmatrix} = \begin{pmatrix} \cos(2t) & -\sin(2t) \\ \sin(2t) & \cos(2t) \end{pmatrix}$$

Differentiate the entries and compare with $\boldsymbol{A}\boldsymbol{\Phi}(t)$.

$$\boldsymbol{\Phi}'(t) = \begin{pmatrix} -2\sin(2t) & -2\cos(2t) \\ 2\cos(2t) & -2\sin(2t) \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} \cos(2t) & -\sin(2t) \\ \sin(2t) & \cos(2t) \end{pmatrix} = \boldsymbol{A}\boldsymbol{\Phi}(t)$$

Fundamental Matrix Inverse

$\boldsymbol{y}' = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)\boldsymbol{y} + \left(\begin{smallmatrix} t \\ 0 \end{smallmatrix}\right)$

The determinant is

$$\det \boldsymbol{\Phi}(t) = \cos^2(2t) + \sin^2(2t) = 1$$

The columns are therefore independent for every t. The inverse is

$$\boldsymbol{\Phi}(t)^{-1} = \begin{pmatrix} \cos(2t) & \sin(2t) \\ -\sin(2t) & \cos(2t) \end{pmatrix}$$

Variation of Parameters

$\boldsymbol{y}' = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)\boldsymbol{y} + \left(\begin{smallmatrix} t \\ 0 \end{smallmatrix}\right)$

Multiply the inverse by the forcing vector.

$$\boldsymbol{\Phi}(s)^{-1}\boldsymbol{f}(s) = \begin{pmatrix} \cos(2s) & \sin(2s) \\ -\sin(2s) & \cos(2s) \end{pmatrix}\begin{pmatrix} s \\ 0 \end{pmatrix} = \begin{pmatrix} s\cos(2s) \\ -s\sin(2s) \end{pmatrix}$$

Integration by Parts

$\mathbf{y}' = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} \mathbf{y} + \begin{pmatrix} t \\ 0 \end{pmatrix}$

In each component, differentiate s and integrate the trigonometric factor.

$$\begin{aligned} \int_0^t s\cos(2s)\,ds &= \left[\frac{s\sin(2s)}{2}\right]_0^t - \frac{1}{2}\int_0^t \sin(2s)\,ds \\ &= \frac{t\sin(2t)}{2} + \frac{\cos(2t)-1}{4} \end{aligned}$$

$$\begin{aligned} \int_0^t -s\sin(2s)\,ds &= \left[\frac{s\cos(2s)}{2}\right]_0^t - \frac{1}{2}\int_0^t \cos(2s)\,ds \\ &= \frac{t\cos(2t)}{2} - \frac{\sin(2t)}{4} \end{aligned}$$

General Solution

$\boldsymbol{y}' = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)\boldsymbol{y} + \left(\begin{smallmatrix} t \\ 0 \end{smallmatrix}\right)$

Collect the two integrals in a vector $\boldsymbol{g}(t)$.

$$\boldsymbol{g}(t) = \int_0^t \boldsymbol{\Phi}(s)^{-1}\boldsymbol{f}(s)\,ds = \begin{pmatrix} t\sin(2t)/2 + (\cos(2t) - 1)/4 \\ t\cos(2t)/2 - \sin(2t)/4 \end{pmatrix}$$

$$\boldsymbol{u}(t) = \boldsymbol{g}(t) + \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

Multiply by $\boldsymbol{\Phi}(t)$ and distribute.

$$\boldsymbol{y}(t) = \boldsymbol{\Phi}(t)\boldsymbol{g}(t) + \boldsymbol{\Phi}(t)\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

General Solution (continued)

$\boldsymbol{y}' = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)\boldsymbol{y} + \left(\begin{smallmatrix} t \\ 0 \end{smallmatrix}\right)$

Expand the matrix product by columns.

$$\begin{aligned}
\boldsymbol{y}(t) &= \boldsymbol{\Phi}(t)\boldsymbol{g}(t) + \begin{pmatrix} \cos(2t) & -\sin(2t) \\ \sin(2t) & \cos(2t) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\
&= \boldsymbol{\Phi}(t)\boldsymbol{g}(t) + c_1 \begin{pmatrix} \cos(2t) \\ \sin(2t) \end{pmatrix} + c_2 \begin{pmatrix} -\sin(2t) \\ \cos(2t) \end{pmatrix} \\
&= \underbrace{\boldsymbol{\Phi}(t)\boldsymbol{g}(t)}_{\text{particular solution}} + \underbrace{c_1\boldsymbol{y}_1(t) + c_2\boldsymbol{y}_2(t)}_{\text{complementary solution}}
\end{aligned}$$

Initial Condition $y(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ $\quad y' = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} y + \begin{pmatrix} t \\ 0 \end{pmatrix}$

$$\begin{aligned} \boldsymbol{y}(0) &= \boldsymbol{\Phi}(0)\boldsymbol{g}(0) + \boldsymbol{\Phi}(0)\boldsymbol{c} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\ &= \boldsymbol{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \end{aligned}$$

Substitute these constants into the general solution.

$$\boxed{\boldsymbol{y}(t) = \boldsymbol{\Phi}(t) \begin{pmatrix} 1 + t\sin(2t)/2 + (\cos(2t) - 1)/4 \\ 2 + t\cos(2t)/2 - \sin(2t)/4 \end{pmatrix}}$$

Verification

$\boldsymbol{y}' = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)\boldsymbol{y} + \left(\begin{smallmatrix} t \\ 0 \end{smallmatrix}\right)$

Write $\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{u}$ with $\boldsymbol{u}(t) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \boldsymbol{g}(t)$. The product rule gives

$$\begin{aligned} \boldsymbol{y}'(t) &= \boldsymbol{\Phi}'(t)\boldsymbol{u}(t) + \boldsymbol{\Phi}(t)\begin{pmatrix} t\cos(2t) \\ -t\sin(2t) \end{pmatrix} \\ &= \boldsymbol{A}\boldsymbol{\Phi}(t)\boldsymbol{u}(t) + \boldsymbol{\Phi}(t)\boldsymbol{\Phi}(t)^{-1}\begin{pmatrix} t \\ 0 \end{pmatrix} \\ &= \boldsymbol{A}\boldsymbol{y}(t) + \begin{pmatrix} t \\ 0 \end{pmatrix} \end{aligned}$$

The solution has the required initial value.

$$\boldsymbol{y}(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$